

Healthcare Clinical Intelligence Platform

Advanced Production-Grade Ingestion Architecture

Version: 1.0

Status: Approved Architecture

Purpose

The ingestion platform transforms raw healthcare knowledge into structured, governed, searchable, and retrievable knowledge assets.

The system is designed for:

- Multi-Tenant SaaS
 - 10M+ Chunks
 - 100K+ Documents
 - 1000+ Daily Uploads
 - Compliance-Ready Operations
 - Low Cost Processing
 - High Retrieval Accuracy
-

Core Design Principles

Principle 1

Retrieval Quality > Model Quality

The quality of retrieval determines answer quality.

Principle 2

Metadata Is A First-Class Citizen

Every chunk must contain rich metadata.

Principle 3

Governance Before Retrieval

No content becomes searchable without approval.

Principle 4

Multimodal By Design

The system supports:

- Text
 - Tables
 - Charts
 - Images
 - Scanned Documents
 - Structured Medical Data
-

Principle 5

Cost Optimization First

Use traditional processing wherever possible.

Use LLMs only when necessary.

High Level Architecture

```
Document Sources
|
Upload Gateway
|
File Validation
|
Document Intelligence Pipeline
|
Metadata Enrichment Pipeline
|
```

```
Chunking Engine
|
Knowledge Graph Extraction
|
Embedding Pipeline
|
Validation Pipeline
|
Approval Workflow
|
Indexing Layer
|
Qdrant + Neo4j + Supabase
```

Supported Sources

Internal Sources

- SOPs
- Guidelines
- Policies
- Treatment Protocols
- Drug Formularies
- Insurance Policies
- Patient Education Material

External Sources

- PubMed
 - WHO
 - FDA
 - CDC
 - NICE
 - RxNorm
 - SNOMED CT
 - ICD-10
-

Supported File Types

Documents

- PDF
- DOCX
- PPTX
- XLSX
- CSV
- TXT

Images

- PNG
- JPG
- TIFF

Structured Data

- JSON
 - XML
 - FHIR
-

Ingestion Services

Upload Service

Responsibilities:

- File Upload
- Presigned URLs
- Virus Scan
- File Validation

Stores original file in S3.

Classification Service

Classifies documents into:

- Research Paper
- Clinical Guideline
- SOP
- Drug Reference

- Insurance Policy
- Lab Report
- Medical Image
- General Document

Output drives chunking strategy.

Parsing Service

Primary Tool:

- Docling

Extracts:

- Text
 - Headings
 - Layout
 - Tables
 - Figures
 - Citations
 - Footnotes
-

OCR Service

Primary Tool:

- PaddleOCR

Used for:

- Scanned PDFs
 - Images
 - Forms
-

Multimodal Service

Primary Tool:

- ColQwen

Used only when needed.

Triggers:

- Charts
- Complex Tables
- Medical Forms
- Layout-heavy Documents

Produces multimodal embeddings.

Metadata Enrichment Pipeline

Objective

Generate retrieval-enhancing metadata.

Extracted Metadata

```
{
  "organization_id": "",
  "department_id": "",
  "knowledge_base_id": "",
  "document_type": "",
  "medical_specialty": "",
  "audience": "",
  "source": "",
  "version": "",
  "approval_status": "",
  "publication_date": "",
  "author": ""
}
```

Medical Metadata

Extract:

- Diseases
- Drugs
- Procedures
- Symptoms
- Treatments

- Guidelines
-

Risk Metadata

Generate:

- Low Risk
 - Medium Risk
 - High Risk
-

Ontology Enrichment

Integrated Standards

- SNOMED CT
 - RxNorm
 - ICD-10
 - LOINC
-

Example

```
Diabetes
|
ICD10:E11

Metformin
|
RxNorm:6809
```

Store mappings inside metadata.

Chunking Architecture

No fixed chunking.

Adaptive Chunking Only.

Chunking Engine

Research Paper Chunker

Structure:

- Abstract
- Introduction
- Methods
- Results
- Discussion
- Conclusion

Chunk by section.

SOP Chunker

Structure:

- Policy
- Procedure
- Exception
- Reference

Chunk by heading.

Guideline Chunker

Parent Child Architecture.

Parent:

Clinical Guideline

Child:

Specific Recommendation

Table Chunker

One table = one chunk.

Includes:

- Header
 - Rows
 - Summary
-

Semantic Chunker

Uses:

- Topic Boundaries
 - Context Changes
 - Section Changes
-

Hierarchical Chunker

Stores:

Document → Section → Subsection → Chunk

Supports hierarchical retrieval.

Graph-Aware Chunker

Chunks linked to:

- Diseases
- Drugs
- Symptoms
- Treatments

Supports GraphRAG.

Chunk Metadata

Every chunk contains:

```
{  
  "chunk_id": "",  
}
```

```
"document_id": "",
"organization_id": "",
"department_id": "",
"knowledge_base_id": "",
"document_version": "",
"roles": [],
"specialty": "",
"source": "",
"risk_level": "",
"entities": []
}
```

Embedding Pipeline

Text Embeddings

Model:

BGE-M3

Image Embeddings

Model:

ColQwen

Table Embeddings

Specialized Table Representation

Stored separately.

Knowledge Graph Pipeline

Extract:

- Disease
- Drug

- Symptom
- Procedure
- Treatment
- Research Paper
- Guideline

Store inside Neo4j.

Versioning Architecture

Version Storage

Never overwrite.

Example:

Guideline v1 Guideline v2 Guideline v3

Active Version

Only latest approved version is active.

Historical Retrieval

Admins can access all versions.

Incremental Reindexing

When new version uploaded:

```
Compare V1 vs V2
  |
Changed Sections
  |
Rechunk
  |
Reembed
```

Do not process unchanged content.

Governance Workflow

States

Draft → Pending Review → Approved → Archived

Retrieval Rule

Only Approved content is searchable.

Approval Authority

- Organization Admin
 - Department Admin
-

Validation Pipeline

Checks:

- Empty Chunks
 - OCR Confidence
 - Duplicate Chunks
 - Metadata Completeness
 - Ontology Mapping Quality
-

Quality Scores

Each document receives:

Parse Score

OCR Score

Metadata Score

Chunk Quality Score

Queue Architecture

Queue

Redis + Celery

Workers

Parser Worker

Metadata Worker

Chunk Worker

Embedding Worker

Graph Worker

Validation Worker

Index Worker

Failure Recovery

Every stage is retryable.

Dead Letter Queue for failures.

No document loss allowed.

Storage Architecture

S3

Stores:

- Raw Files
 - Parsed Files
 - Images
 - Tables
 - Versions
-

Supabase

Stores:

- Metadata
 - Jobs
 - Audit Logs
 - Versions
-

Qdrant

Stores:

- Text Embeddings
 - Image Embeddings
 - Table Embeddings
-

Neo4j

Stores:

- Medical Graph
-

Observability

Track:

- Upload Latency
 - Parse Latency
 - OCR Latency
 - Chunk Count
 - Embedding Cost
 - Graph Nodes Created
 - Processing Failures
 - Approval Rate
-

Cost Optimization Strategy

Never Use LLM For

- OCR
 - Parsing
 - Basic Metadata
 - Duplicate Detection
-

Use LLM Only For

- Metadata Enrichment
 - Medical Classification
 - Complex Entity Extraction
-

Cache

- Embeddings
 - Metadata
 - PubMed Results
 - Ontology Mappings
-

Incremental Processing

Only changed content is reprocessed.

Production Success Metrics

Target:

- 10M+ Chunks
 - 100K+ Documents
 - <60 Seconds Processing Time Per Average Document
 - 95% Metadata Completeness
 - 90% Chunk Quality Score
 - 99% Successful Processing Rate
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Final Principle

The ingestion system is not a document uploader.

It is a Healthcare Knowledge Intelligence Pipeline responsible for converting raw information into governed, searchable, auditable, and retrieval-optimized knowledge assets.